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State-of-charge dynamic balancing strategy for distributed energy storage system in DC shipboard microgrid

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ABSTRACT

In this paper, a State-of-Charge (SoC) dynamic balancing control strategy considering system communication failure and energy storage capacity difference is proposed to reach the SoC balancing and proper current sharing for distributed energy storage units (DESUs) in DC shipboard microgrid. Firstly, the distributed equalization algorithm is applied in the DESU's controller when the system communication is normal, each DESU exchanges their SoC information through low bandwidth communication (LBC) and dynamically adjusts the droop coefficient according to the change of their SoC, so as to realize the SoC dynamic balance of the energy storage system. Secondly, the DESU's controller is switched to the decentralized equalization algorithm with communication failure, each DESU can reach SoC balancing and load current reasonable distribution only according to the local information. After that, a DC bus voltage compensation method is introduced to solve the problem of bus voltage deviation caused by the proposed SoC equalization. Finally, the simulation model comprised of three DESUs is implemented and the results show that the proposed control strategy can make each DESU reach SoC equilibrium state quickly and effectively reduce the bus voltage deviation under the normal and failure communication conditions.

1. Introduction

With the gradual exhaustion of global fossil fuels and the increasing environmental pollution problems, energy-saving and emission-reduction technologies in the shipping industry enjoy attracted widespread attentions [1–3]. At present, new energy sources such as solar, wind and hydrogen energy have unique advantages in energy-saving and emission-reduction, which can effectively alleviate the energy crisis [4–6]. In the field of ship transportation, with the gradual maturity of new power electronic and power battery technologies, as well as the continuous decline of power electronic equipment costs, the promotion of new energy ships in the shipping market has shown a surge. New energy ships generally use all-electric propulsion, and their power system can be equivalent to an islanding microgrid which composed of renewable energy power generation system, energy storage system, AC/DC converters, various loads, and protection devices [7]. In contrast to AC microgrid, the DC microgrid do not have the problems caused by reactive power, harmonics and synchronization [8], its operation mode is more flexible and the power supply reliability is more excellent. Therefore, the application of DC microgrid in new energy ship power system has been paid more and more attention.

However, unlike the terrestrial DC microgrid, the DC shipboard microgrid has been operating in an independent mode and cannot exchange power with the AC grid. In addition, ship-borne wind power, photovoltaic (PV) and other new energy power generation systems are not only affected by natural conditions, but also by various complex navigation conditions of ships, their output power fluctuate more frequently. Hence, in order to ensure the safe navigation of ships, it is indispensable to introduce energy storage devices to stabilize the power fluctuations caused by new energy power generation, so as to maintain the energy balance of the microgrid and the stability of the bus voltage [9].

To meet the large-capacity requirements of the DC shipboard microgrid system, energy storage modules are usually connected to the DC bus in parallel, thus forming a distributed energy storage system (DESS) [10]. Nevertheless, due to the unreasonable load current sharing of each DESU during the charging and discharging process, there are differences in their SoCs. If the DESUs with unbalanced SoCs continue to work, it is easy to cause some DESUs to overcharge and deep discharge, thus shortening their lifetimes. Furthermore, it may even affect the

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stable operation of the DC shipboard microgrid owing to the early withdrawal of certain DESUs from work. Therefore, it is necessary to research a dynamic balance control strategy for shipboard energy storage units with considering their SoCs, so as to realize the autonomous and reasonable sharing of output current between each paralleled converter, and promote the SoC of each DESU to reach equilibrium quickly and stably.

According to the degree of dependence on the communication network, the load current sharing control methods for parallel DESUs can be divided into centralized, distributed and decentralized [11]. Among them, the centralized control method is simple and easy to implement, while it has the disadvantages of high communication cost, unfavorable system redundancy, and the risk of single-point failure [12–14]. Currently, in the research literature of the past few years, both distributed control and decentralized control belong to peer-to-peer control, which can achieve the goal of autonomous load current sharing without high-frequency communication, and meet the requirements of distributed system access. Therefore, the droop control can be used as an ideal current sharing control method to be applied to the DESS of the DC shipboard microgrid.

In the field of decentralized control, some researchers have developed various energy storage SoC equalization algorithms without underlying communication [15,16]. In [17–19], a SoC balancing method for DESS is proposed, which detects the optimal n value to achieve SoC equalization and power sharing by setting the droop coefficient to be inversely proportional to the n-th power of the SoC. This method only requires to control the power exponent n to effectively achieve SoC equilibrium, but only the discharge situation is considered. When the SoC is close to zero, the droop coefficient tends to infinity, which has certain limitations. The description of energy storage charging is supplemented in [20], whereas the inherent defects of the algorithm itself still exist. In [21], on the basis of the SoC equalization algorithm proposed in [17–20], the SoC-based droop control strategy is combined with the voltage regulation method based on the directed graph, so as to achieve power sharing and bus voltage recovery, meanwhile, an observer based on the directed graph is designed to enhance the tracking function of the reference current. This strategy can address the problems of power balance, bus voltage stability and system disturbance. However, the power sharing situation with different capacity of the energy storage unit is not considered, and the proposed control strategy is too complicated to be applied to engineering practice. In [22], a novel decentralized control method is proposed, which can realize SoC dynamic balance based on voltage and current double closed-loop control. Nonetheless, this method has lower SoC equalization accuracy. In [23], a decentralized strategy based on fuzzy control is proposed, which can dynamically modify the droop coefficient according to the SoC to maintain the energy storage balance and realize the bus voltage recovery. This method does not rely on communication links and has strong robustness. Unfortunately, the formulation of fuzzy reasoning rules has strong subjectivity. Moreover, the fuzzy rules proposed in the literature are too simple to guarantee the SoC equalization accuracy.

In the field of distributed control, most of the existing researches transmit SoC information between adjacent DESUs through LBC. In [24], a distributed energy storage management strategy is proposed, which introduced an auxiliary controller to calculate the average SoC of the DESS when the communication is normal, and the droop coefficient is dynamically adjusted by combining the energy storage SoC and the average SoC with the exponential function. When communication failure occurs, the traditional droop control is switched to maintain the normal system operation. This method can realize the SoC balance and the load current sharing when the system communication is normal. However, the deviation of the bus voltage and the difference in energy storage capacity on the SoC equalization effect are not considered. Furthermore, when the system is switched to traditional droop control, the SoC equalization capability becomes invalid. On the basis, the power exponent n is introduced in [25] to improve the resolution of the energy storage SoC, thereby speeding up the search for the optimal droop curve, so that the energy storage SoC can converge to an equilibrium state. In [26], a consensus algorithm is used to make the state information only transmitted between adjacent microgrids, and the droop coefficient can be adjusted in real time according to the SoC changes. This method can relieve the system communication pressure and achieve the goal of load current sharing. In [27], a SoC-based virtual power rating method is proposed to achieve accurate load current sharing. In fact, none of the abovementioned literatures considers the situation of communication failure and the difference in energy storage capacity. In [28], the threshold of SoC is divided to coordinate the energy management of each parallel energy storage unit, and the voltage regulator is optimized to weaken the influence of communication delay on bus voltage recovery. However, the SoC equalization speed of the proposed strategy is relatively slow.

Through the analysis of the aforementioned related literatures, it can be known that the distributed control method can achieve a relatively ideal SoC equalization effect by using LBC network to transmit adjacent information or global average information. However, the disadvantage is that every DESU need to maintain communication contact, and the algorithm may be invalid when the communication fails. On the contrary, the decentralized control method does not need to establish the underlying communication structure for the DESUs, as long as the own state information is transmitted in the local converter to complete the SoC balance target, and has a pleasurable “plug-and-play” performance. Nevertheless, employing this method would result in poor SoC balance accuracy, and it is difficult to achieve accurate load current sharing. In a word, most of the past researches only considered individual centralized, decentralized or distributed control methods to achieve the SoC balance, which could not in accordance with the demands of various operating modes of DC shipboard microgrid in practical applications. Consequently, on the basis of previous researches, and comprehensively considering the advantages and disadvantages of distributed control and decentralized control methods, a novel SoC dynamic balancing strategy of DESS is proposed for the DC shipboard microgrid system. The main contributions of this paper are as follows.

1) When the system communication is normal, the energy storage controller employs a distributed control method, which uses LBC to transmit SoC information in each DESUs. Particularly, the droop coefficient is associated with the DESUs’ own SoC and the global average SoC to achieve dynamic SoC balancing. This method effectively realizes the balanced dispatch of local energy and the reasonable load current sharing in the DESS. Compared with the existing SoC equalization methods, the SoC convergence speed and accuracy of the distributed control method proposed in this paper can be greatly improved.
2) Considering the possible communication failure in the practical applications, a decentralized control algorithm is designed, which does not need to establish the underlying communication for each DESU. With this method, each DESU only needs to transmit the local state information to achieve the SoC balance goal. Unfortunately, the SoC convergence effect of decentralized control method is slightly worse than that of distributed control method, so it is only used when communication fails.
3) In other to solve the problem of bus voltage deviation caused by SoC-based droop control, a secondary compensation method is introduced to restore the bus voltage to a given value for ensuring the safe and stable operation of the DC shipboard microgrid.

The rest of this paper is structured as follows. Section 2 presents the DC shipboard microgrid system configuration. In Section 3, the basic knowledge of traditional droop control is summarized, the SoC dynamic balance strategy for the DC shipboard microgrid DESS is proposed, and the secondary compensation method to quickly restore the bus voltage is introduced. Then, the system equivalent model and its stability analysis

are performed in Section 4. The simulation results under different working conditions are reported in Section 5. Finally, the conclusion is drawn in Section 6.

2. DC shipboard microgrid structure

The actual engineering ship discussed in this paper is an all-electrical solar ship. The main ship power sources are the electrical energy generated by the shipboard PV power station and DESS. The PV array can be installed on the open deck and the outer wall of the superstructure. The function of DESUs is to store the excess electricity generated by the PV array when there is sufficient light, and to supply power to the load during periods of insufficient light such as night or rainy days. Therefore, the performance of DESUs directly affects the PV power generation system reliability. In actual applications, the corresponding controller must be configured to control its charging/discharging, so that each DESU can determine the power distribution according to its own SoC, thereby prolong its lifetime.

The difficulty in the design of all-electric propulsion ships lies in the realization of DC networking technology [29]. The ship power system including new energy generation mostly adopts DC network, and its typical structure is shown in Fig. 1. It is mainly composed of PV power generation system, DESS, AC and DC power equipment, and main propulsion motors, etc. Each module in the Fig. 1 is connected to the DC bus through a converter, and the red arrow represents the direction of energy flow.

Due to the real-time power fluctuation of PV power generation side and load side, each DESU is required to adjust its charging and discharging states to compensate for the power shortage of microgrid or absorb surplus power, so as to maintain the stability of DC bus voltage.

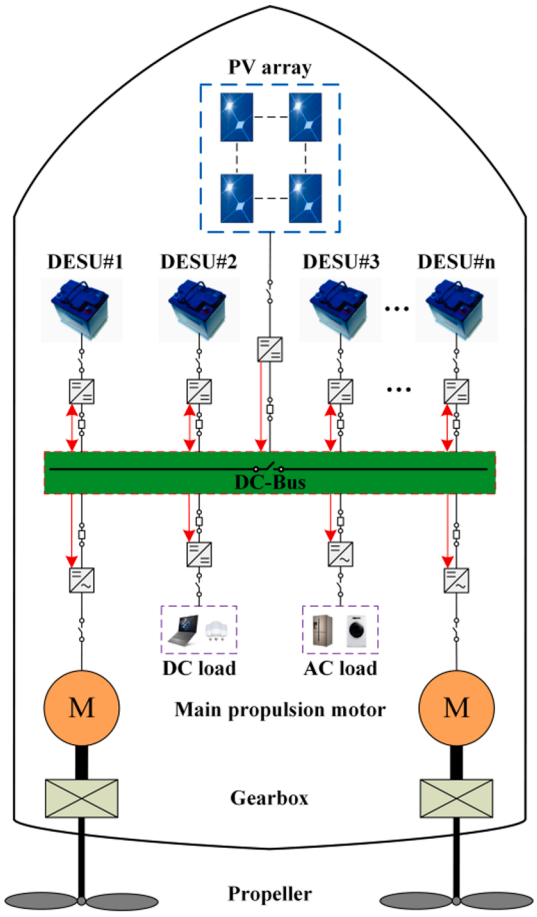


Fig. 1. DC shipboard microgrid structure.

In other to simplify the analysis, it is assumed that each DESU is operated in the SoC safe area (i.e., 20%<SoC < 90%), and the DESUs will stop running when their SoCs are higher than the upper limit or lower than the lower limit. Each DESU in the microgrid can operate in charge/discharge mode or idle mode. The specific mode depends on the positive or negative difference between the power generated by the PV system and the power consumed by the load. The PV power generation system runs in maximum power point tracking (MPPT) mode. The power balance relationship can be shown as

$$P_{\mathrm{PV}} + P_{\mathrm{DESS}} - P_{\mathrm{Loss}} = P_{\mathrm{Load}} \tag{1}$$

where P_{pv} is the power generated by the PV power generation system, P_{DESS} is the power compensated or absorbed by the DESS, P_{Loss} is the power loss of line resistance and power electronic converters, P_{Load} is the total power consumed by the load.

In the process of system power balance, it is indispensable to ensure that each DESU can reasonably allocate the responsible power according to its SoC, so that the DESUs with high SoCs can absorb less power during charging and release more power during discharge. On the contrary, the DESUs with low SoCs can absorb more power during charging and release less power during discharge. As a result, the SoC balance and the accurate current sharing of each DESU are realized.

3. SoC dynamic balancing strategy of DESS

3.1. Basic theory of traditional droop control

The droop control of DC shipboard microgrid is mainly realized by introducing a virtual resistance into the front end of the voltage and current double closed loop to adjust the output current between the parallel converters. The principle is shown in Fig. 2.

The expression of droop control in DC microgrid can be shown as

$$U_{\mathrm{ref}i} = U_{\mathrm{dc}}^{*} - I_{\mathrm{dc}i}R_{\mathrm{d}i} \tag{2}$$

where $U_{\mathrm{ref}i}$ is the output voltage reference value of the i-th converter,U_{dc}^{*} is the rated value of the DC bus voltage, $I_{\mathrm{dc}i}$ and $R_{\mathrm{d}i}$ are the output current and droop coefficient of the i-th converter, respectively.

The equivalent circuit diagram with two paralleled DESUs in DC microgrid is shown in Fig. 3, where $R_{\mathrm{line}i}$ is the line impedance, R_{Load} is the load resistance, U_{PCC} is the voltage at the common connection point of the converter, i.e., the actual value of DC bus voltage.

Combining the droop control formula with Kirchhoff 's voltage law can be obtained that

$$\begin{cases} U_{\mathrm{PCC}} = U_{\mathrm{dc}}^{*} - I_{\mathrm{dc1}}R_{\mathrm{d1}} - I_{\mathrm{dc1}}R_{\mathrm{line1}} \\ U_{\mathrm{PCC}} = U_{\mathrm{dc}}^{*} - I_{\mathrm{dc2}}R_{\mathrm{d2}} - I_{\mathrm{dc2}}R_{\mathrm{line2}} \end{cases} \tag{3}$$

From (3), it can be derived as

$$\frac{I_{\mathrm{dc1}}}{I_{\mathrm{dc2}}} = \frac{R_{\mathrm{d2}} + R_{\mathrm{line2}}}{R_{\mathrm{d1}} + R_{\mathrm{line1}}} = \frac{R_{\mathrm{d2}}}{R_{\mathrm{d1}}} + \frac{R_{\mathrm{line2}} - \frac{R_{\mathrm{d2}}}{R_{\mathrm{d1}}}R_{\mathrm{line1}}}{R_{\mathrm{d1}} + R_{\mathrm{line1}}} \tag{4}$$

To ensure accurate current sharing between parallel converters, it yields

$$\frac{R_{\mathrm{d1}}}{R_{\mathrm{d2}}} = \frac{R_{\mathrm{line1}}}{R_{\mathrm{line2}}} \tag{5}$$

Therefore, (4) can be rewritten as

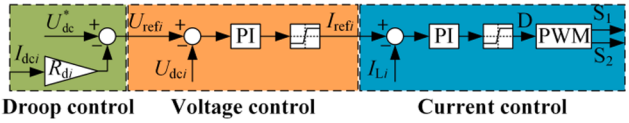


Fig. 2. Traditional droop control principle.

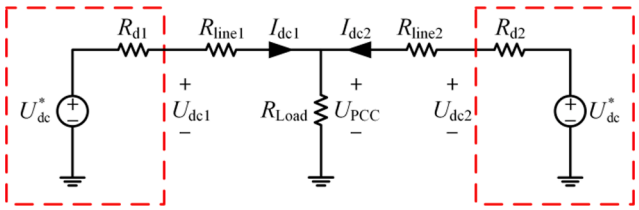


Fig. 3. Equivalent circuit diagram of two DESUs in parallel.

$$\frac{I_{dc1}}{I_{dc2}} = \frac{R_{d2}}{R_{d1}} \tag{6}$$

From (6), it can be seen that the output current of the parallel converter is inversely proportional to its corresponding droop coefficient.

In addition, from (2), it can be known that the droop control will bring about a significant deviation in the output voltage of the converter, and the deviation value of the bus voltage can be express as

$$\Delta U_{dci} = I_{dci} R_{di} \tag{7}$$

The choice of the droop coefficient depends on the maximum allowable deviation of the bus voltage and the maximum allowable output current of the converter. Too large or too small droop coefficient will affect the system stability, so its value is restricted as

$$0 < R_{di} \leqslant \frac{\Delta U_{dcmax}}{I_{dcmax}} \tag{8}$$

where ΔU_{dcmax} is the maximum allowable deviation of the DC bus voltage, I_{dcmax} is the maximum allowable output current of the converter.

3.2. Dynamic SoC balancing control strategy

According to the above theoretical analysis, it can be known that when traditional droop control is employed, the droop coefficient of each converter is fixed. If the droop coefficients or the line impedance are inconsistent, the equivalent output impedance of each converter is not equal, which would inevitably cause the converter output current to be inconsistent, thereby forming a circulating current between the converters and affecting the stable operation of the DC microgrid. In addition, due to the special characteristics of the DESUs, overcharge or deep discharge will shorten their lifetime. Therefore, it is indispensable to control each DESU to determine the output power according to their SoCs, which can prevent some DESUs from overcharging or deep discharging. However, the traditional droop control cannot adjust the droop coefficient in real time. For this reason, a dynamic SoC balancing strategy for DESS based on adaptive droop control is proposed. Considering the normal communication and communication failure of the system, the distributed equalization algorithm and decentralized equalization algorithm are designed in the strategy respectively.

The Control structure of energy storage unit is shown in Fig. 4. When the system communication is normal, the distributed equalization algorithm is used in the DESUs, in which the auxiliary controller is used to calculate the SoC average value and the information is transmitted through the LBC line. Other information is only transmitted in the local converter. When the system communication fails, the decentralized equalization algorithm is employed in each DESU, which does not need to establish the underlying communication, and only relies on local information to achieve the SoC equalization goal. In the whole process, the priority of each DESU is always equal. Even if a certain DESU fails in the system, the normal operation of other DESUs would not be affected. Therefore, the DC shipboard microgrid system would have superior redundancy and reliability.

3.2.1. Distributed equalization algorithm

The SoC estimation of the energy storage usually employs the

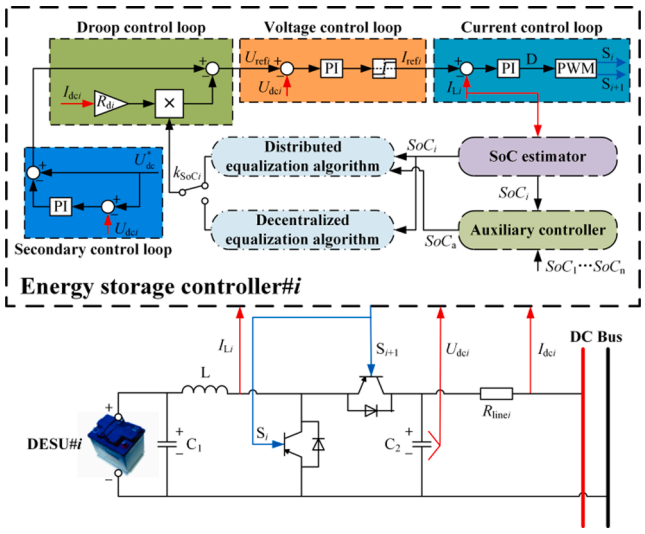


Fig. 4. Control structure of energy storage unit.

Coulomb counting method [30], as shown in

$$SoC_i(t) = SoC_i(0) - \frac{\eta_i}{C_{bati}} \int_0^t I_{bati} \mathrm{d}t \tag{9}$$

where $SoC_i(t)$ and $SoC_i(0)$ are the SoC values of the i-th DESU at the current time and the initial time respectively, η_i is the charge/discharge efficiency of the i-th converter, C_{bati} is the rated capacity of the i-th DESU, I_{bati} is the output current of the i-th DESU.

Derivation of (9) can get the SoC change rate, it yields that

$$SoC_i'(t) = -\frac{\eta_i I_{bati}}{C_{bati}} \tag{10}$$

Under ideal circumstances, substituting (2) into (10) can be obtained

$$SoC_i'(t) = -\frac{\eta_i (U^*_{dc} - U_{refi})}{C_{bati} R_{di}} \tag{11}$$

Each DESU is connected in parallel to the DC bus, they have the same U_{ref}, and the rated capacity C_{bati} of each DESU has been determined during manufacture. Hence, it can be seen from (11) that the change rate of the SoC can be adjusted by changing the bus reference voltage U_{ref} or the droop coefficient R_{di}, thereby indirectly adjusting the power absorbed/released by each DESU, and finally promoting the gradual convergence of their SoCs. If the SoC is balanced by changing the bus reference voltage, the bus voltage would be further deviated and it is difficult to recover to the given value. Therefore, the two control algorithms mentioned in this paper both dynamically adjust the droop coefficient to achieve SoC balance.

When the system communication is normal, the energy storage controller employs the distributed equalization algorithm, and the droop control formula is modified as follows

$$U_{refi} = U^*_{dc} - I_{dci} R_{SoCi} \tag{12}$$

where R_{SoCi} is the improved droop coefficient, which can be express as

$$R_{SoCi} = R_{di} k_{SoCi} \tag{13}$$

where R_{di} is the initial droop coefficient, k_{SoCi} is the dynamic compensation factor for droop coefficient.

The key of SoC balance lies in the design of the dynamic compensation factor k_{SoCi}. The dynamic compensation factor expression of distributed equalization algorithm is designed as follows.

$$k_{\mathrm{SoC}i} = \begin{cases} \dfrac{SoC_a}{C_{\mathrm{bat}i}SoC_i} \mathrm{e}^{\left[\frac{\rho}{\alpha\left|SoC_i^n - SoC_a^n\right|+\beta}(SoC_i^n - SoC_a^n)\right]}, & if\ I_{\mathrm{dc}i} < 0 \\ \dfrac{SoC_a}{C_{\mathrm{bat}i}SoC_i} \mathrm{e}^{\left[-\frac{\rho}{\alpha\left|SoC_i^n - SoC_a^n\right|+\beta}(SoC_i^n - SoC_a^n)\right]}, & if\ I_{\mathrm{dc}i} > 0 \end{cases} \tag{14}$$

where SoC_a is the dynamic average SoC of parallel DESUs, SoC_i is the dynamic SoC value of the i-th DESU, ρ is the equalization acceleration factor, and its value has a greater impact on the SoC equilibrium speed, α and β are the equalization speed factor and equalization precision factor respectively, which are mainly used to control the initial speed and final accuracy of SoC equalization, n is the convergence factor, and its value affects the SoC convergence speed and convergence effect.

From (12) and (13), the output current of the i-th converter is

$$I_{\mathrm{dc}i} = \frac{U_{\mathrm{dc}}^* - U_{\mathrm{PCC}}}{R_{\mathrm{d}i}k_{\mathrm{SoC}i}} \tag{15}$$

To simplify the derivation process, suppose that:

$$F_i = \frac{\rho}{\alpha\left|SoC_i^n - SoC_a^n\right| + \beta} \tag{16}$$

Taking the discharge process of two parallel DESUs as an example, combining (9), (14), (15) and (16), it is derived

$$SoC_i(t) = SoC_i(0) - \int_0^t \frac{\eta_i SoC_i(t)\left(U_{\mathrm{dc}}^* - U_{\mathrm{PCC}}\right)}{R_{\mathrm{d}i}SoC_a(t)\mathrm{e}^{F_i(t)\left[SoC_a^n(t) - SoC_i^n(t)\right]}}\mathrm{d}t \tag{17}$$

Assuming $R_{\mathrm{d}1} = R_{\mathrm{d}2} = R_0$ and $\eta_1 = \eta_2 = \eta$, from (17), the SoC difference between the two DESUs can be obtained

$$\begin{aligned} \Delta SoC_{12}(t) &= SoC_1(t) - SoC_2(t) \\ &= SoC_1(0) - SoC_2(0) - \\ &\int_0^t \frac{\eta SoC_1(t)(U_{\mathrm{dc}}^* - U_{\mathrm{PCC}})}{R_0 SoC_a(t)\mathrm{e}^{F_1(t)\left[SoC_a^n(t) - SoC_1^n(t)\right]}}\mathrm{d}t + \\ &\int_0^t \frac{\eta SoC_2(t)(U_{\mathrm{dc}}^* - U_{\mathrm{PCC}})}{R_0 SoC_a(t)\mathrm{e}^{F_2(t)\left[SoC_a^n(t) - SoC_2^n(t)\right]}}\mathrm{d}t \end{aligned} \tag{18}$$

Then, the derivative of (18) can be written as

$$\begin{aligned} \Delta SoC_{12}'(t) = &\frac{\eta SoC_2(t)(U_{\mathrm{dc}}^* - U_{\mathrm{PCC}})}{R_0 SoC_a(t)\mathrm{e}^{F_2(t)\left[SoC_a^n(t) - SoC_2^n(t)\right]}} - \\ &\frac{\eta SoC_1(t)(U_{\mathrm{dc}}^* - U_{\mathrm{PCC}})}{R_0 SoC_a(t)\mathrm{e}^{F_1(t)\left[SoC_a^n(t) - SoC_1^n(t)\right]}} \end{aligned} \tag{19}$$

The first-order expansion of Taylor series of exponential function e^x can be approximately expressed as

$$\mathrm{e}^x = 1 + x \tag{20}$$

By combining (19) and (20), it yields

$$\begin{aligned} \Delta SoC_{12}'(t) = &\frac{\eta\cdot(U_{\mathrm{dc}}^* - U_{\mathrm{PCC}})}{R_0\cdot SoC_a(t)}\cdot \\ &\left[\frac{SoC_2(t)}{1 + F_2(t)\left[SoC_a^n(t) - SoC_2^n(t)\right]} - \right. \\ &\left. \frac{SoC_1(t)}{1 + F_1(t)\left[SoC_a^n(t) - SoC_1^n(t)\right]}\right] \end{aligned} \tag{21}$$

It can be seen from (18) and (21) that when $SoC_1(0) < SoC_2(0)$, that is, $\Delta SoC_{12}(t) < 0$, and $\Delta SoC_{12}'(t) > 0$ at this time, thus $\Delta SoC_{12}(t)$ approaches zero from the negative direction of the vertical axis of the two-dimensional coordinate system over time. That is to say, the SoC difference between two parallel DESUs gradually decreases, as shown by the blue dashed line in Fig. 5. Similarly, when $SoC_1(0) > SoC_2(0)$, that is, $\Delta SoC_{12}(t) > 0$, and $\Delta SoC_{12}'(t) < 0$ at this time, so $\Delta SoC_{12}(t)$ approaches zero from the positive direction of the vertical axis of the two-dimensional coordinate system over time. In other words, the SoC difference between the two parallel DESUs gradually decreases, as shown by the red dashed line in Fig. 5. The charging process of DESUs can be deduced by analogy.

The theoretical analysis above shows that the proposed distributed equalization algorithm can realize the adaptive adjustment of the droop coefficient, so as to gradually reduce the SoC difference between each DESU, and finally achieve the dynamic SoC balancing. When the DESUs are charged, the DESU with lower SoC would have smaller droop coefficient, larger charging current and faster SoC rise rate. In contrast, the DESU with higher SoC would have larger droop coefficient, smaller charge current and slower rise rate. When the DESUs are discharged, the DESU with lower SoC would have larger droop coefficient, smaller discharge current, and slower SoC drop rate. On the contrary, the DESU with higher SoC would have smaller droop coefficient and larger discharge current, and their SoC decline rate are faster. According to this SoC equalization characteristic, the SoCs of parallel DESUs can be guided to gradually converge, thereby the energy of each DESU can be balanced. Each DESU can reach the SoC critical value during the charge/discharge process simultaneously, which avoids some DESUs from overcharging/deep discharging or ending operation early, so as to protect the energy storage service life and improve energy utilization.

According to (13) and (14), assuming $R_{\mathrm{d}i} = 3$, $SoC_a = 0.5$, $\alpha = 4$, $\beta = 0.01$, the three-dimensional surface diagram of droop coefficient $R_{\mathrm{SoC}i}$ can be drawn when different values of n and ρ are selected in the charge/discharge process of DESUs, which is shown in Fig. 6. It can be seen from Fig. 6 that when DESUs are discharged, if $SoC_i < SoC_a$, $R_{\mathrm{SoC}i}$ would increase with the increasing of ρ, and decrease with the increasing of n. If $SoC_i > SoC_a$, $R_{\mathrm{SoC}i}$ would decrease with the increasing of ρ and n. When DESUs are charged, if $SoC_i < SoC_a$, $R_{\mathrm{SoC}i}$ would decrease with the increasing of ρ and n. If $SoC_i > SoC_a$, $R_{\mathrm{SoC}i}$ would increase with the increasing of ρ, and decrease with the increasing of n.

From the above analysis, it can be known that when the difference between SoC_i and SoC_a is large, the influence of ρ and n is greater. Therefore, the values of ρ and n cannot be too large, and should be selected according to actual conditions.

What's more, the values of α and β also have a greater impact on the SoC convergence process. Assuming $R_{\mathrm{d}i} = 3$, $\rho = 5$, $n = 2$, the two-dimensional graph of $R_{\mathrm{SoC}i}$ during the charging/discharging process of DESUs can be drawn when α and β are different, which is shown in Fig. 7. It can be concluded from Fig. 7 that when DESUs are discharged and $\Delta SoC < 0$, the DESU has less electricity, its droop coefficient should be larger and the discharge current responsible should be smaller. At this

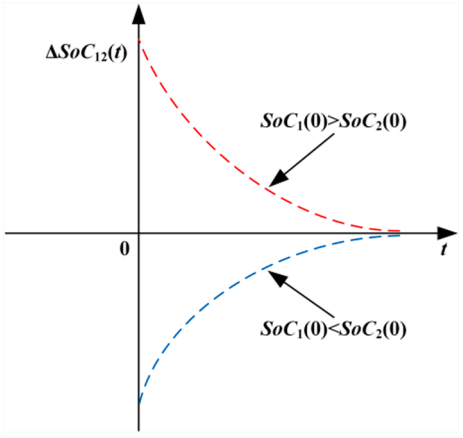


Fig. 5. The convergence of $\Delta SoC_{12}(t)$-curve.

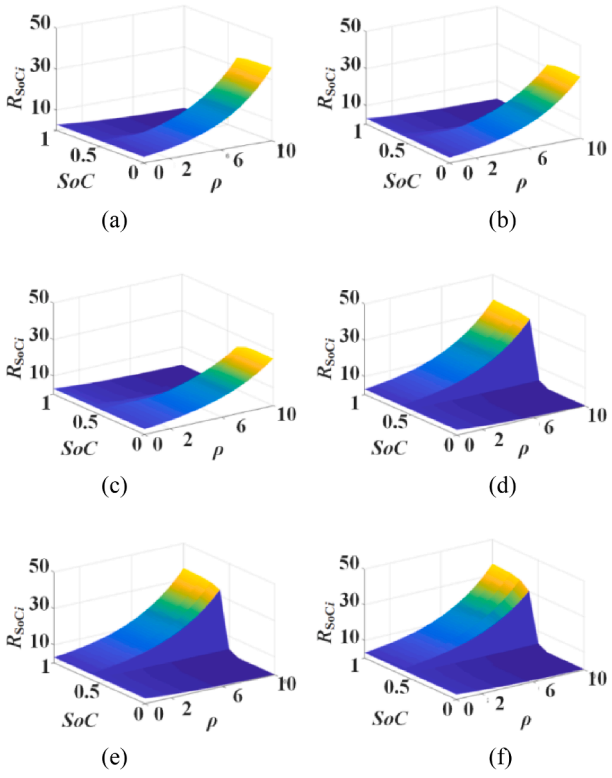


Fig. 6. R_{SoCi} three-dimensional surface graph with different values of n and ρ. (a) $I_{dc} > 0$, $n = 2$. (b) $I_{dc} > 0$, $n = 4$. (c) $I_{dc} > 0$, $n = 6$. (d) $I_{dc} < 0$, $n = 2$. (e) $I_{dc} < 0$, $n = 4$. (f) $I_{dc} < 0$, $n = 6$.

time, if $\Delta SoC > 0$, the DESU stores more electricity, thus its droop coefficient should be smaller and the discharge current should be larger. When DESUs are charging, if $\Delta SoC < 0$, then the DESU stores less electricity, consequently its droop coefficient should be smaller and the charge current should be larger. At this time, if $\Delta SoC > 0$, the DESU stores more electricity, accordingly its droop coefficient should be larger and the charging current should be smaller.

From the aforementioned analysis results, it can be seen that α mainly affects the speed of the initial stage of SoC equalization, and β mainly affects the accuracy of the final stage of SoC equalization. Therefore, it is indispensable to select appropriate values of α and β according to actual requirements to obtain more favorable SoC equalization speed and accuracy.

According to (14), (16) and $\Delta SoC_i^n = SoC_i^n - SoC_a^n$, the selection rules for parameters ρ, n, α and β are given as follows:

(1) When the SoC difference of each DESU is small, ΔSoC_i^n tends to zero. From (16) we know that $\alpha|\Delta SoC_i^n| \approx 0$. At this time, in order to obtain a faster equalization speed, $\beta < 1$ is required, then (16) can be simplified as $F_i = \rho/\beta > 0$, so the SoC equalization speed and accuracy are mainly determined by ρ and β with the SoC difference is small.

(2) When the SoC difference of each DESU is large, in order to avoid the obvious influence of β on the SoC equalization and obtain a faster equalization speed, $\alpha|\Delta SoC_i^n| \gg \beta$ is required, so the order of magnitude of α should be at least one level larger than β, then (16) can be simplified as $F_i = \rho/\alpha|\Delta SoC_i^n| > 0$, so the SoC equalization speed is mainly determined by ρ, α and n with the SoC difference is large.

Considering the theoretical analysis, the selection rules of each parameter and a large number of simulation results, $\rho = 5$, $n = 2$, $\alpha = 4$, $\beta = 0.01$ are selected in this paper.

3.2.2. Decentralized equalization algorithm

When a communication failure occurs in the DESS, the communication connection between each DESU is lost. If the distributed equalization algorithm continues to be applied, the SoC would not be able to converge. Therefore, a distributed equalization algorithm is proposed, which can promote the DESUs to keep a superior SoC balance effect continuously with the system has a communication failure.

The droop coefficient dynamic compensation factor design of the proposed decentralized equalization algorithm is as follows

$$k_{SoCi} = \begin{cases} \dfrac{SoC_{ref}}{C_{bati} SoC_i} e^{\left[\frac{\rho\left(SoC_i^n - SoC_{ref}^n\right)}{SoC_{ref}^n}\right]}, & if\ I_{dci} < 0 \\ \dfrac{SoC_{ref}}{C_{bati} SoC_i} e^{\left[-\frac{\rho\left(SoC_i^n - SoC_{ref}^n\right)}{SoC_{ref}^n}\right]}, & if\ I_{dci} > 0 \end{cases} \tag{22}$$

$$SoC_{ref} = \frac{SoC_H + SoC_L}{2} \tag{23}$$

where SoC_H and SoC_L are the SoC charging upper limit and discharging lower limit respectively, and all DESUs are equal. In this paper, $SoC_H = 90\%$, $SoC_L = 20\%$. SoC_{ref} is the reference value of SoC equilibrium, which is determined by the upper and lower limits of SoC charge/discharge. Here, $SoC_{ref} = 55\%$ is selected.

3.3. DC bus voltage recovery strategy

When droop control is used between parallel converters, it would result in the bus voltage to deviate from a given value. In other words, the bus voltage rises when the DESUs is charging, and drops when the DESUs is discharging. In order to solve this problem, a bus voltage recovery strategy is introduced to compensate the bus voltage and the bus voltage finally stable near the given value.

This strategy is applicable to both normal and fault communication, and the realization of bus voltage recovery only needs PI controller. The

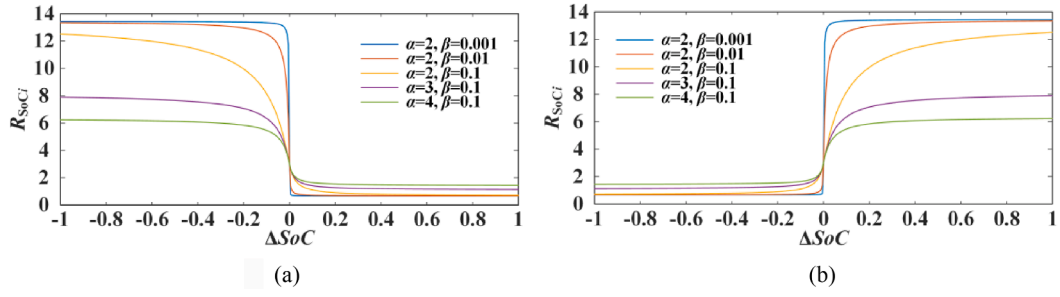


Fig. 7. Two-dimensional curves of R_{SoCi} with different values of α and β. (a) $I_{dc} > 0$. (b) $I_{dc} < 0$.

bus voltage compensation formula can be designed as

$$\delta = \left(K_P + \frac{K_I}{s}\right)\left(U_{dc}^* - U_{bus}\right) \tag{24}$$

where δ is the secondary compensation amount of bus voltage, K_P and K_I are the proportional coefficients and the integral coefficient of the PI controller in the bus voltage compensation loop, respectively.

Finally, the complete improved droop control formula is as follows

$$U_{refi} = U_{dc}^* - I_{dci}R_{di}k_{SoCi} + \delta \tag{25}$$

4. Stability analysis of proposed method

In this section, the island DC shipboard microgrid which involved two DESUs with equal capacity is taking as an example to analyze the stability of the SoC dynamic balancing strategy.

The equivalent model of the i-th DESU is shown in Fig. 8. The DC voltage closed-loop transfer function $G_c(s) = 1/(\tau s + 1)$ [31], where τ is the time constant under the action of the switch, and it generally takes a small value. Therefore, the transfer function can be approximated as $G_c(s) = 1$. Furthermore, the SoC estimation can be equivalent to an inertial link [32], and ω_c is the cut-off frequency of the low-pass filter.

From Fig. 8, it yields that

$$\begin{cases} U_{dci}(s) = U_{dc}^* - R_{SoC}\dfrac{1}{s+1}\dfrac{\omega_c}{s+\omega_c}\left(\alpha_i U_{dci} - \gamma U_{dcj}\right) \\ U_{dcj}(s) = U_{dc}^* - R_{SoC}\dfrac{1}{s+1}\dfrac{\omega_c}{s+\omega_c}\left(\alpha_j U_{dcj} - \gamma U_{dci}\right) \end{cases} \tag{26}$$

where

$$\begin{cases} \alpha_i = \dfrac{R_{linej} + R_{Load}}{R_{linei}R_{linej} + R_{linei}R_{Load} + R_{linej}R_{Load}} \\ \alpha_j = \dfrac{R_{linei} + R_{Load}}{R_{linei}R_{linej} + R_{linei}R_{Load} + R_{linej}R_{Load}} \\ \gamma = \dfrac{R_{Load}}{R_{linei}R_{linej} + R_{linei}R_{Load} + R_{linej}R_{Load}} \end{cases} \tag{27}$$

in which, R_{linei}, R_{linej} and R_{Load} are the line impedance of the i-th DESUs, the line impedance of the j-th DESUs, and the load impedance, respectively. Combining (12), (26) and (27), the system characteristic equation can be obtained

$$As^4 + Bs^3 + Cs^2 + Ds + E = 0 \tag{28}$$

where

$$\begin{cases} A = 1 \\ B = 2 + 2\omega_c \\ C = 1 + 4\omega_c + \omega_c^2 - (\alpha_i + \alpha_j)R_{SoC} \\ D = 2\omega_c + 2\omega_c^2 - (\alpha_i + \alpha_j)(1 + \omega_c)\omega_c R_{SoC} \\ E = \omega_c^2 - (\alpha_i + \alpha_j)\omega_c^2 R_{SoC} + (\omega_c R_{SoC})^2(\alpha_i\alpha_j - \gamma^2) \end{cases} \tag{29}$$

The characteristic equation of the control system is required to meet the Routh stability criterion. In other words, the coefficients must satisfy $(A, B, C, D, E) > 0$, then the value of the droop coefficient can be determined by the Routh stability criterion and (8). According to the system parameters which is shown in Table 1, the root locus diagram of the droop coefficient R_{SoCi} and cut-off frequency ω_c can be drawn, as shown in Fig. 9.

From Fig. 9(a) and (b), it can be seen that the dominant poles λ_4 and λ_3 are basically not affected by the changes of droop coefficient and cut-off frequency. The conjugate poles λ_1 and λ_2 gradually move away from the imaginary axis as the droop coefficient and cut-off frequency increase, which shows the system stability can be guaranteed.

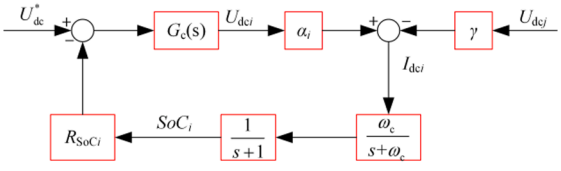


Fig. 8. Equivalent model of stability analysis.

5. Simulation results

In order to verify the effectiveness and feasibility of the proposed control strategy, a simulation model as shown in Fig. 10 is built through MATLAB/Simulink for various case analyses, the simulation results are compared with reference [24]. The system simulation parameters are shown in Table 2.

5.1. Case I: Performance of the proposed method with stable charge/discharge process

This case verifies the SoC equalization effect of the stable charging/discharging process of the DESS when the communication is normal. During the charging process, the initial SoCs of the three DESUs are 50%, 40%, and 30% respectively, the power generated by the PV array is constant at 3.1 kW, and the load power is 1 kW. During the discharging process, the initial SoCs of the three DESUs are 80%, 70%, and 60% respectively, the power generated by the PV array is also constant at 3.1 kW, and the load power is 5 kW. In addition, the rated capacity of each DESU is set to 5 Ah to quickly observe the equalization effect of the proposed algorithm.

Fig. 11 and Fig. 12 show the simulation results of the stable charging and discharging process of the DESS respectively. From Fig. 11 (a), (c) and Fig. 12 (a), (c), it can be seen that when the method proposed in this paper is used, the SoC of each DESU converges stably, and the output current of each converter gradually tends to be consistent. At about $t = 20$ s, the SoC balance and the output current sharing of each converter are realized, and the current sharing speed and accuracy are relatively high. When the method proposed in [24] is used, as shown in Fig. 11(b), (d) and Fig. 12(b), (d), the SoC and load current of each DESU can also gradually converge, but the convergence speed is slow, and the SoC balance and load current sharing goals are not achieved within the set simulation time range. What's more serious is that as SoC differences gradually become smaller, the method proposed in [24] will weaken the ability to identify SoC differences, which will cause the SoC equalization speed to slow down, and it is easy to cause some DESUs to exit work early and affect the system stable operation. Additionally, as shown in Fig. 11(e), (f) and Fig, 12(e), (f), the method proposed in this paper can maintain the bus voltage near 400 V, while the method proposed in [24] will cause the bus voltage to deviate from the given value.

The above analysis verifies that the proposed method can make the SoC converge quickly and stably during the stable charging/discharging process of DESS, realize the reasonable load current distribution, and ensure that the bus voltage is maintained within the allowable range.

Table 1
System parameters.

Item	Symbol	Value	Unit
Load resistance	R_{Load}	160	Ω
Line impedance of the i-th DESU	R_{linei}	1	Ω
Line impedance of the j-th DESU	R_{linej}	0.5	Ω
Droop coefficient	R_{SoCi}	0.1 ~ 3	Ω
Cut-off frequency of low-pass filter	ω_c	35 ~ 100	rad/s

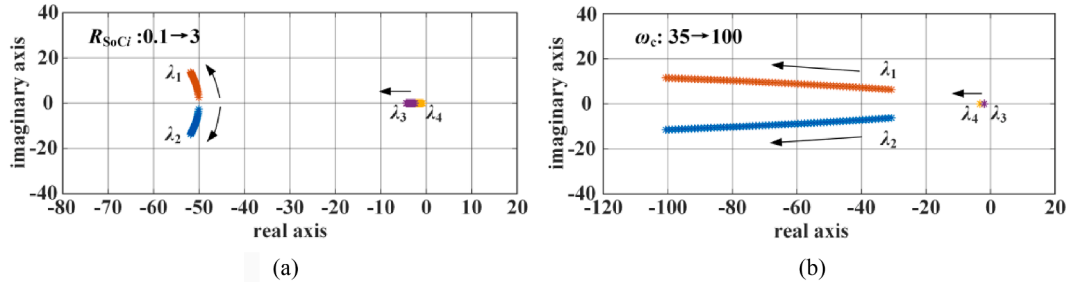


Fig. 9. Root locus with different parameters. (a) $\omega_c = 50$ rad/s, R_{SoCi} increases. (b) $R_{SoCi} = 0.5$, ω_c increases.

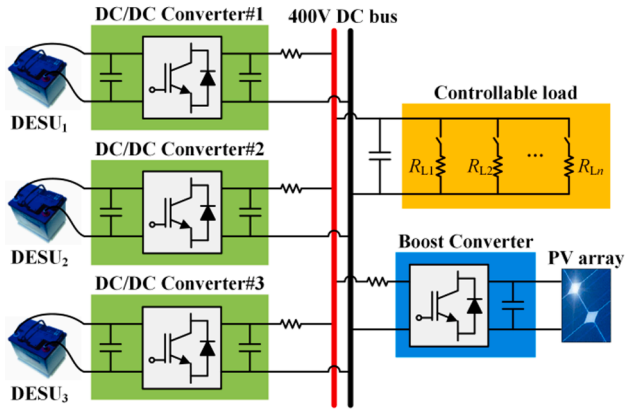


Fig. 10. Simulation model.

Table 2
System simulation parameters.

Item	Symbol	Value	Unit
Nominal Voltage of the three DESUs	U_{bat1}	200	V
	U_{bat2}	200	V
	U_{bat3}	200	V
Input side capacitance of DC/DC converter	C_1	300	μF
output side capacitance of DC/DC converter	C_2	500	μF
Inductance of DC/DC converter	L	2	mH
Rated bus voltage	U_{dc}^*	400	V
Line impedance of the three converters	R_{line1}	0.5	Ω
	R_{line2}	0.8	Ω
	R_{line3}	1.1	Ω
Switching frequency	f_k	20	kHz
Cut-off frequency	ω_c	50	rad/s

5.2. Case II: Performance of the proposed method with sudden load changes and PV output power fluctuation

This case mainly analyzes the dynamic performance of the system with the load power or PV output power changes suddenly when the system communication is normal. Assuming that the initial SoC of the three DESUs are 60%, 50%, and 40% respectively, the initial load power is 5 kW, which is suddenly reduced to 1 kW at $t = 5$ s, and suddenly increased to 4 kW at $t = 15$ s. The initial power generated by the PV array is 3.1 kW, which is suddenly increased to 5.1 kW at $t = 25$ s, and suddenly reduced to 2.1 kW at $t = 35$ s. The rated capacity of each DESU is 5Ah.

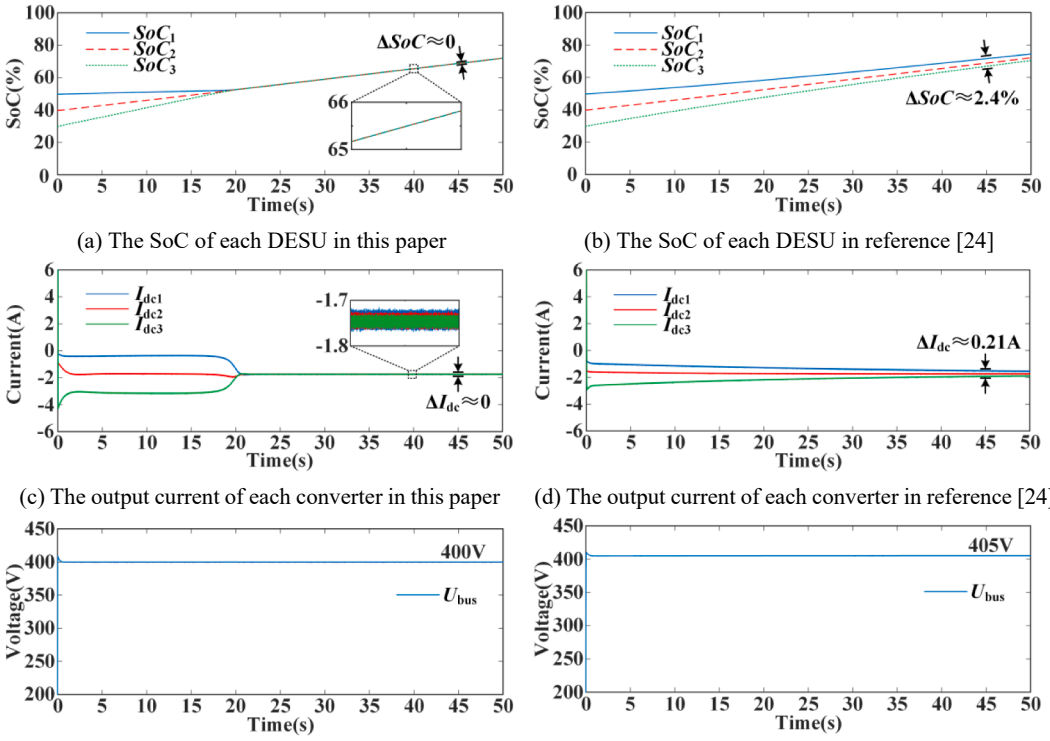
Fig. 13 shows the simulation results when the PV power and load power change suddenly. During $t = 0 \sim 5$ s, the power generated by the PV array is not enough to supply the load consumption, each DESU works in the discharge mode to compensate for the insufficient power, and the SoC and load current gradually converge. At $t = 5$ s, the load power is suddenly reduced to 1 kW, and the power generated by the PV array is greater than the load consumption, then each DESU switches from the discharging mode to the charging mode to absorb the surplus power of the system. At $t = 15$ s, the load power suddenly increases to 4 kW, and each DESU switches to the discharge mode again to provide insufficient power. At $t = 25$ s, the power generated by the PV array suddenly increases to 5.1 kW, and each DESU works in charging mode to absorb the surplus power. At $t = 35$ s, the power generated by the PV array is suddenly reduced to 2.1 kW, and each DESU switches to the discharge mode to maintain the system power balance. In the entire simulation process, when the method proposed in this paper is used, SoC balance and load current sharing are achieved at about $t = 20$ s. When the load power or PV power changes, the bus voltage fluctuates, but the bus voltage can be restored to 400 V under the control of the proposed method, and both voltage and current have better dynamic performance in the transient process of power fluctuation. When using the method proposed in [24], the SoC balance and load current sharing cannot be achieved within the set simulation time, and the bus voltage has a large deviation with the power fluctuates and cannot be maintained at a given value.

The above analysis verifies that the proposed distributed equalization algorithm has faster equalization speed and accuracy, and the dynamic effect of charge and discharge switching of each DESU is excellent, and it can maintain a high bus voltage level.

5.3. Case III: Performance of the proposed method with the 3-th DESU exiting operation and the rated capacity of each DESU differs

This case mainly analyzes the SoC balance of the DESS with the 3-th DESU exits operation due to a failure when the communication is normal and the rated capacity of each DESU is different. Assuming that the initial SoC of the three DESUs is 80%, the power generated by the PV array is constant at 3.1 kW, and the load power is 4 kW. Before $t = 15$ s, the traditional droop control method is used, and the distributed equalization algorithm is switched at $t = 15$ s. At $t = 25$ s, the 3-th DESU quit operation due to a failure. The rated capacity of each DESU is 3Ah, 5Ah and 7Ah respectively.

When the method proposed in this paper is used, as shown in Fig. 14 (a) and (c), the droop control is first adopted before $t = 15$ s. At this time, the droop coefficient of each converter is equal, so the load current can be shared. Since the droop coefficients of the converters are equal at this time, the load current can be shared, but the influence of the SoC on the system is not considered, so the SoC of each DESU gradually diverges. At $t = 15$ s, the energy storage control method is switched to the distributed equalization algorithm, after which the load current is redistributed, and the SoC gradually converges. At $t = 25$ s, the 3-th DESU quit operation due to a fault, and the remaining DESU's SoC can still converge stably, and achieve equilibrium at about $t = 27$ s, and the load current is finally distributed accurately in a ratio of 3:5, which is in line with the expected control objectives. When the method proposed in [24] is used, as shown in Fig. 14(b) and (d), since this method does not consider the difference in the rated capacity of each DESU, the SoC of each DESU still diverges when the droop control method is switched to the SoC equalization algorithm. Therefore, the remaining DESU cannot achieve SoC balance



(a) The SoC of each DESU in this paper

(b) The SoC of each DESU in reference [24]

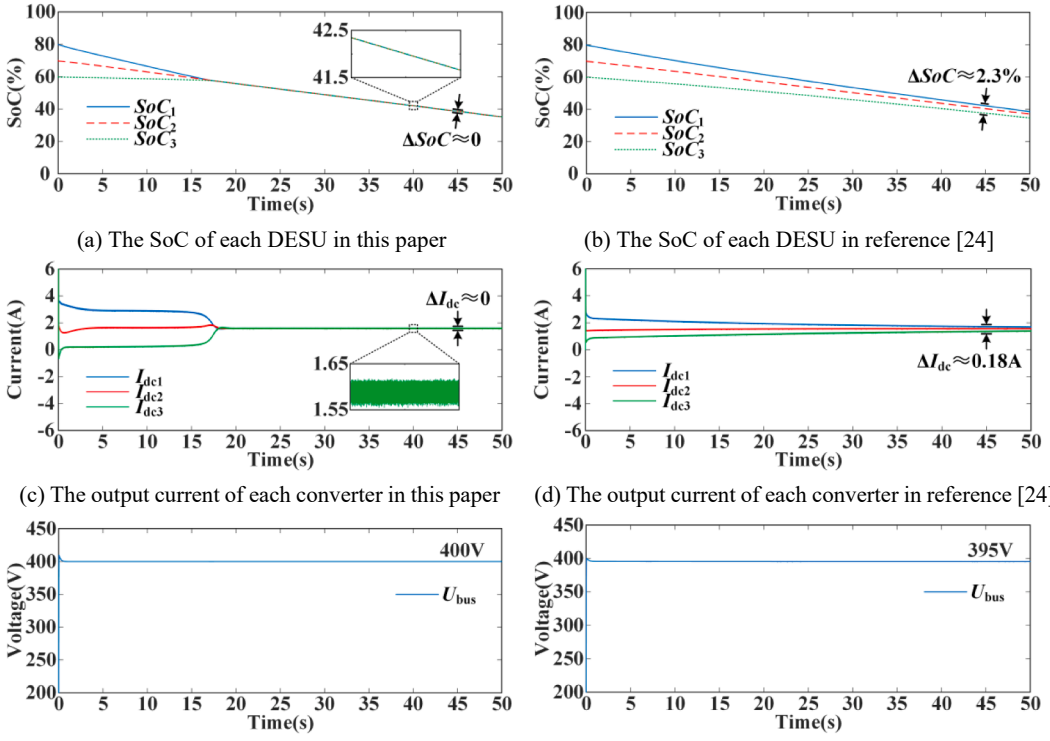
(c) The output current of each converter in this paper

(d) The output current of each converter in reference [24]

(e) DC bus voltage in this paper

(f) DC bus voltage in reference [24]

Fig. 11. The simulation results of case I during charging.



(a) The SoC of each DESU in this paper

(b) The SoC of each DESU in reference [24]

(c) The output current of each converter in this paper

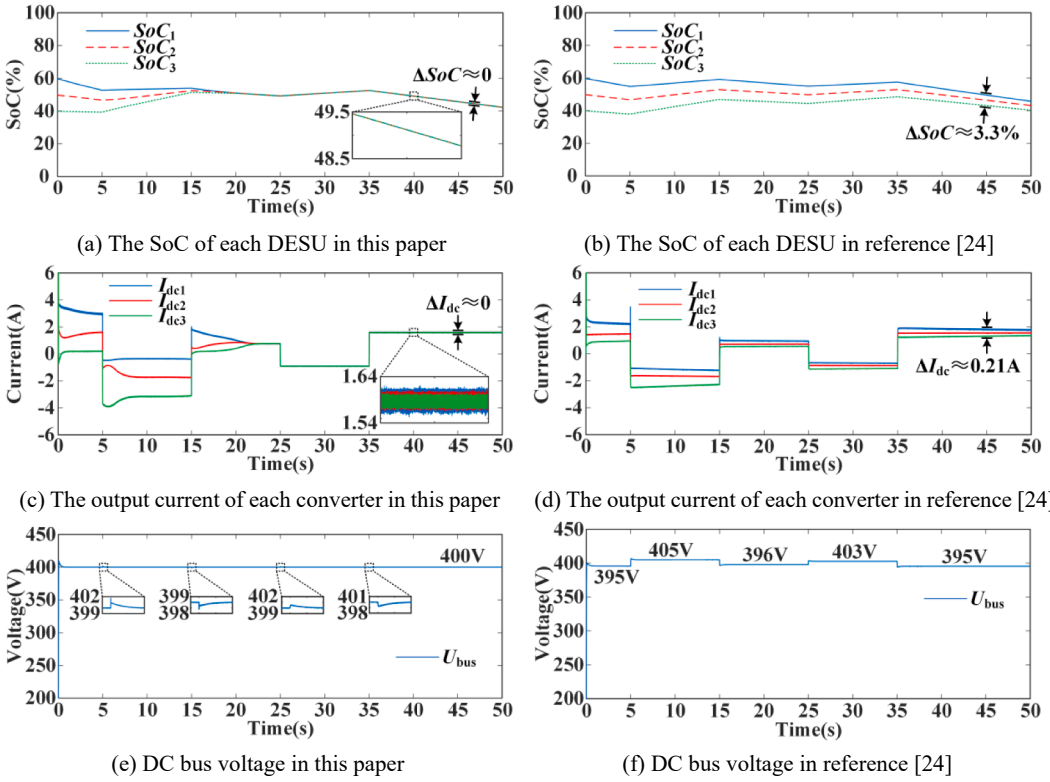
(d) The output current of each converter in reference [24]

(e) DC bus voltage in this paper

(f) DC bus voltage in reference [24]

Fig. 12. The simulation results of case I during discharging.

with the 3-th DESU exits operation, and the load current cannot be distributed proportionally. In addition, as shown in Fig. 14(e) and (f), the bus voltage fluctuates with the algorithm is switched and the 3-th DESU exits operation. However, the method proposed in this paper can restore and stabilize the bus voltage at 400 V, while the bus voltage deviates from the given value to a certain extent when the method



(a) The SoC of each DESU in this paper

(b) The SoC of each DESU in reference [24]

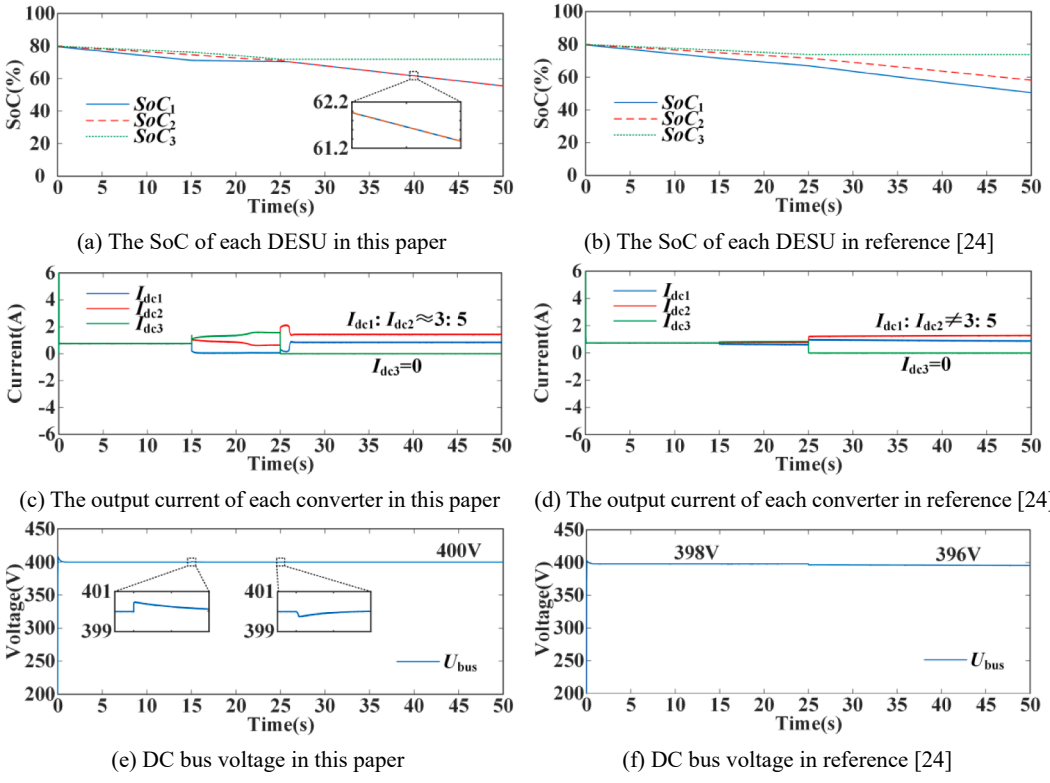
(c) The output current of each converter in this paper

(d) The output current of each converter in reference [24]

(e) DC bus voltage in this paper

(f) DC bus voltage in reference [24]

Fig. 13. The simulation results of case II.



(a) The SoC of each DESU in this paper

(b) The SoC of each DESU in reference [24]

(c) The output current of each converter in this paper

(d) The output current of each converter in reference [24]

(e) DC bus voltage in this paper

(f) DC bus voltage in reference [24]

Fig. 14. The simulation results of case III.

proposed in [24] is used.

Through the above analysis, it is verified that when a certain DESUs exits work due to unexpected circumstances and the rated capacity of each DESU differs, the proposed distributed equalization algorithm can still stably realize the SoC equalization of the remaining DESU and the accurate load current distribution in proportion to the rated capacity of

the energy storage, and keep the bus voltage stable.

5.4. Case IV: Performance of the proposed method with a communication failure occurs

This case mainly analyzes the control effect of the decentralized equalization algorithm when the system communication fails. Assuming that the initial SoCs of the three DESUs are 80%, 70%, and 60% respectively, the power generated by the PV array is constant at 3.1 kW, and the load power is 5 kW. The rated capacity of each DESU is 5Ah. When t = 10 s, the system communication fails, and the system communication recovers when t = 40 s.

When the method proposed in this paper is used, as shown in Fig. 15 (a) and (c), the system communication is normal at the beginning, and the SoC and load current of each DESU gradually converge under the control of the distributed equalization algorithm. At t = 10 s, the system communication fails, the energy storage control algorithm is switched to the decentralized equalization algorithm, each converter does not need to communicate, and the DESU's SoC can still continue to converge without affecting the system normal operation. At t = 40 s, the system communication resumes and the energy storage control algorithm switches back to the distributed equalization algorithm. It can be seen from the figure that in order to quickly equalize the SoC of each DESU, the output current of each converter is redistributed at this time. Finally, the SoC of each DESU reaches equilibrium around t = 43 s. When the method proposed in [24] is adopted, as shown in Fig. 15(b) and (d), it can only be switched to the traditional droop control to maintain the system normal operation with the system communication fails, it will cause some DESUs to exit early and affect the system stability. In addition, as shown in Fig. 15(e) and (f), during the transient process of switching between the two control algorithms, the bus voltage fluctuates, but it can quickly restore stability under the control of the proposed method. However, the method in [24] will cause the bus voltage to deviate from the given value, which is not conducive to the system stable operation.

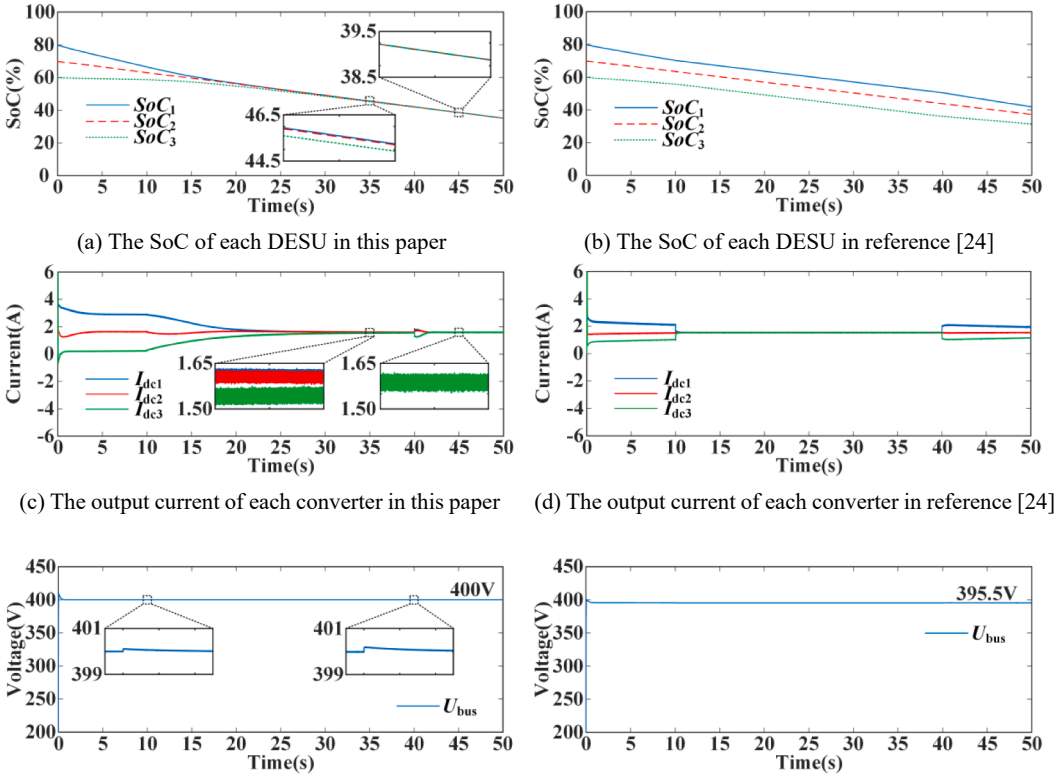
The above analysis verifies that the proposed decentralized equalization algorithm can still maintain the system normal operation and make the SoC of each DESU continue to converge with the system has a communication failure. However, it is not difficult to find through the comparison of the partial enlarged images in Fig. 15(a) and (c) that although the proposed decentralized equalization algorithm can finally achieve SoC equalization, its equalization speed and accuracy are significantly lower than that of the distributed equalization algorithm. Therefore, the decentralized equalization algorithm can only be used as a solution for system communication failures.

6. Conclusion

To solve the problems of unbalanced SoC and unreasonable load current distribution among DESUs of shipborne DC microgrid, a SoC dynamic equilibrium control strategy that takes into account system communication failures and energy storage capacity differences was proposed.

1) When the system communication was normal, a distributed equalization was employed, in which each DESU converged quickly and stably, and obtained a higher SoC equalization speed and accuracy.
2) When the system communication failed, a decentralized equalization algorithm was switched to control the DESS, and each DESU could achieve the SoC balance goal without communication contact.
3) A secondary compensation method for bus voltage was introduced, which solved the problem that the bus voltage deviates from a given value caused by the droop coefficient, and the bus voltage recovery effect was excellent.

Finally, a simulation model for the DESS in DC shipboard microgrid



(a) The SoC of each DESU in this paper

(b) The SoC of each DESU in reference [24]

(c) The output current of each converter in this paper

(d) The output current of each converter in reference [24]

(e) DC bus voltage in this paper

(f) DC bus voltage in reference [24]

Fig. 15. The simulation results of case IV.

was established, which proved the rationality and effectiveness of the proposed control strategy. In the future, it is indispensable to carry out experimental case studies to further verify the performance of the proposed control strategy through actual deployment in real scenarios.

CRediT authorship contribution statement

Yuji Zeng: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft. **Qinjin Zhang:** Validation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Yancheng Liu:** Resources, Supervision, Project administration, Funding acquisition. **Xuzhou Zhuang:** Writing - review & editing. **Long Che:** Writing - review & editing. **Miao Niu:** Writing - review & editing. **Xianglu Zheng:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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